

Gradient Boosting Classifier

Person	Age	Salary	Experience	Bought
P1	22	30k	1	0
P2	25	35k	2	0
P3	40	80k	10	1
P4	45	90k	12	1

Step 1 : Initial prediction

$$\text{Initial probability} = \frac{\text{positive samples}}{\text{negative samples}}$$

$$p = 2/4 = 0.5$$

convert to log odds $\log\left(\frac{p}{1-p}\right)$

$$\log \frac{0.5}{0.5} = 0$$

$$F_0(x) = 0 \quad (\text{Score})$$

Convert score to probability

$$\frac{1}{1+e^{-x}} = 0.5$$

Person	Actual	Initial probability
P1	0	0.5
P2	0	0.5
P3	1	0.5
P4	1	0.5

Step2: Calculate gradient
 Gradient $y-p$

Person	Actual (y)	Probability (P)	(y-P) Gradient
P1	0	0.5	-0.5
P2	0	0.5	-0.5
P3	1	0.5	+0.5
P4	1	0.5	+0.5

Intuition of gradient

Negative
 positive

Probability too high
 Probability too low

Step3: Train first tree

Person	Gradient
P1	-0.5
P2	-0.5
P3	+0.5
P4	+0.5

Suppose best split = Salary < 50k
 Yes No

Left P1, P2, Gradients = -0.5, -0.5
 -0.5 (Avg)

Right P3, P4, Gradients = +0.5, +0.5
 +0.5 (Avg)

Tree learned $\rightarrow$ low sal decrease probability
 High sal increase probability

Step 4: Update Score

$$F(x) = F(x) + \eta T(x)$$

$$\eta = 0.1$$

Old score $\Rightarrow 0$

New score $\Rightarrow 0 + 0.1(-0.5)$

P_1, P_2 -0.05

P_3, P_4 $+0.05$

Convert back to probability

P_1, P_2 $\frac{1}{1+e^{-x}} = \sigma(0.05) \approx 0.487$

P_3, P_4 $= \sigma(0.05) \approx 0.512$

New probabilities

Person	Actual	New probability
P_1	0	0.487
P_2	0	0.487
P_3	1	0.512
P_4	1	0.512

Repeat the process of calculating new gradient and Train new tree